



US 20230079468A1

(19) **United States**
(12) **Patent Application Publication** (10) **Pub. No.: US 2023/0079468 A1**
WATSON et al. (43) **Pub. Date: Mar. 16, 2023**

(54) **MONITORING COMPLIANCE USING
VENOUS REFILL DETECTION**

Publication Classification

(71) Applicant: **KPR U.S., LLC**, Mansfield, MA (US)

(72) Inventors: **Kristin L. WATSON**, Lithia, FL (US);
Mark A. VESS, Hanson, MA (US)

(21) Appl. No.: **17/934,982**

(22) Filed: **Sep. 23, 2022**

(51) **Int. Cl.**
A61B 5/00 (2006.01)
A61B 5/026 (2006.01)
A61H 9/00 (2006.01)
G16Z 99/00 (2006.01)

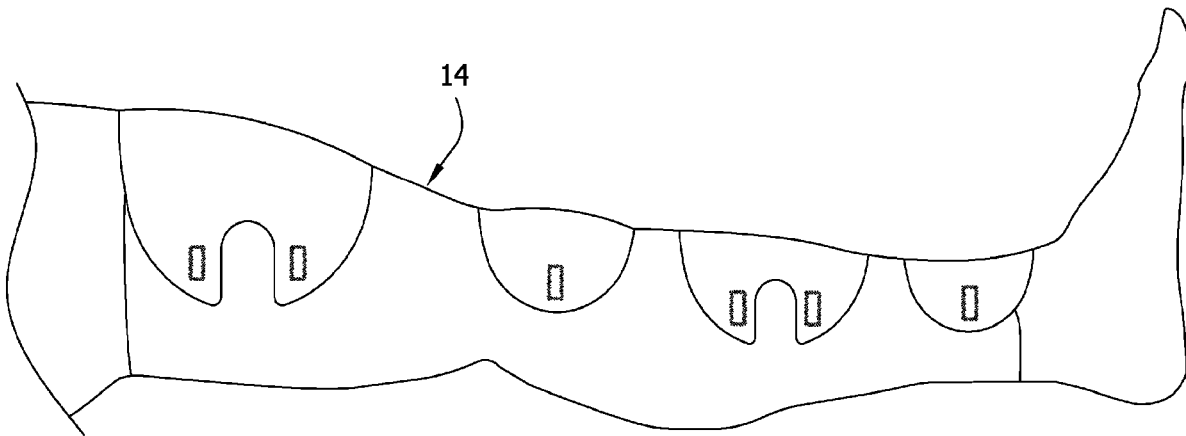
(52) **U.S. Cl.**
CPC *A61B 5/4833* (2013.01); *A61B 5/026*
(2013.01); *A61B 5/6802* (2013.01); *A61B*
5/6828 (2013.01); *A61B 5/7275* (2013.01);
A61H 9/0078 (2013.01); *G16Z 99/00*
(2019.02); *A61B 5/022* (2013.01)

Related U.S. Application Data

(63) Continuation of application No. 16/435,244, filed on Jun. 7, 2019, now abandoned, which is a continuation of application No. 15/452,971, filed on Mar. 8, 2017, now Pat. No. 10,314,531, which is a continuation of application No. 12/894,826, filed on Sep. 30, 2010, now abandoned.

(57) **ABSTRACT**

Monitoring patient compliance with a compression therapy regimen. Venous Refill Time (VRT) is monitored via a pressure sensor in a bladder of a compression system. A controller of the compression system correlates the monitored VRT to a predetermined threshold to determine whether the patient is using the compression system.



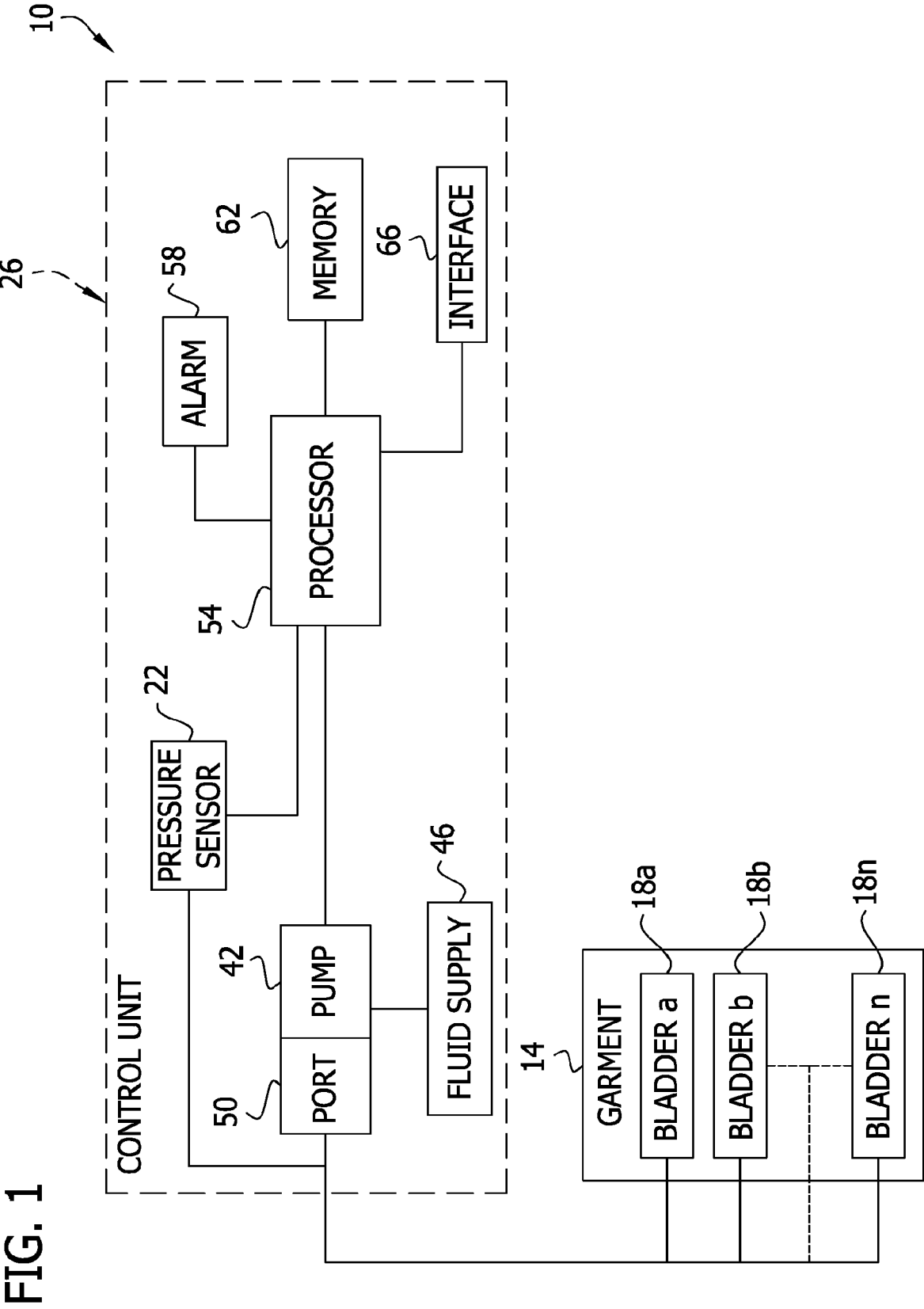


FIG. 2A

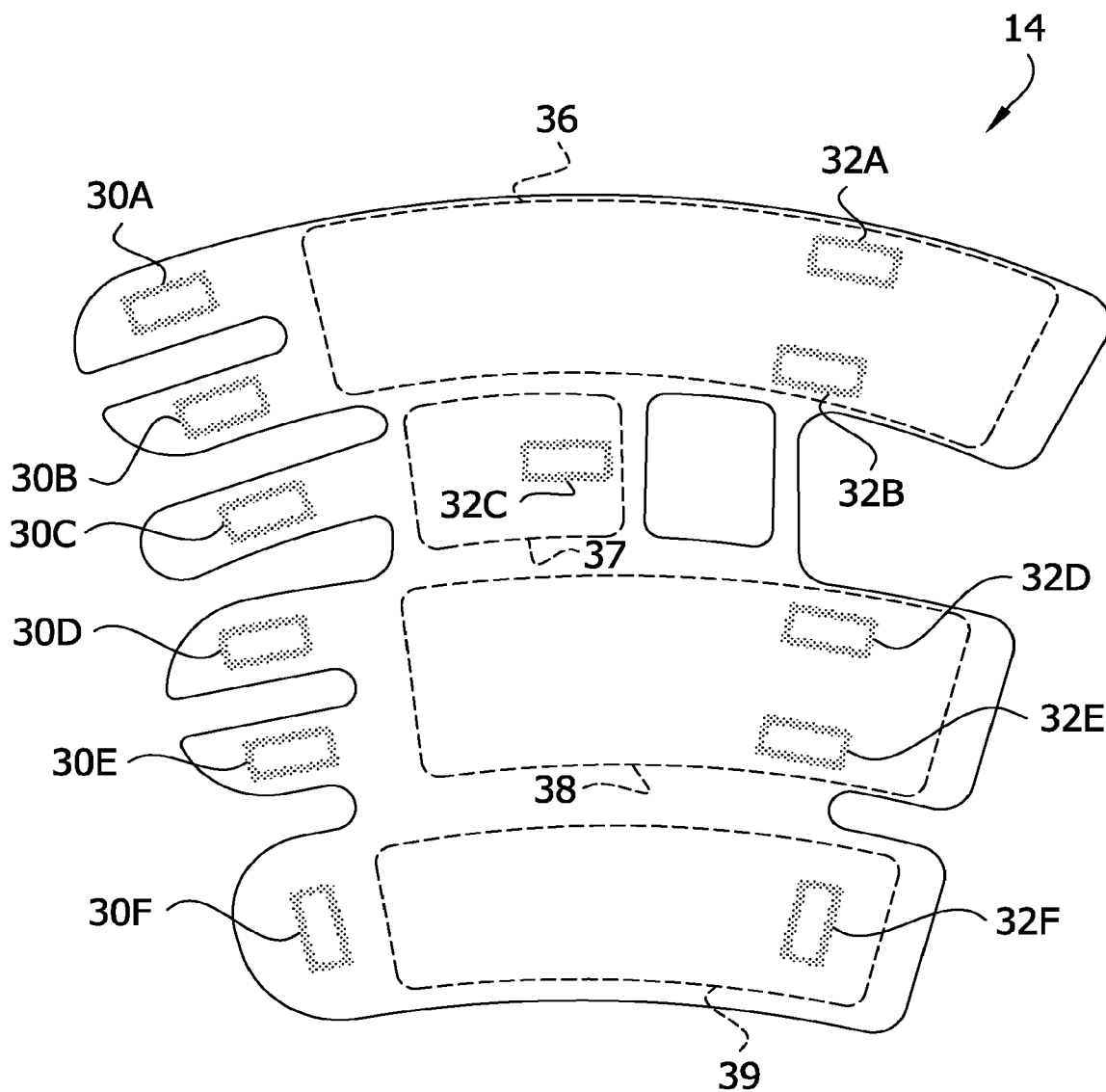


FIG. 2B

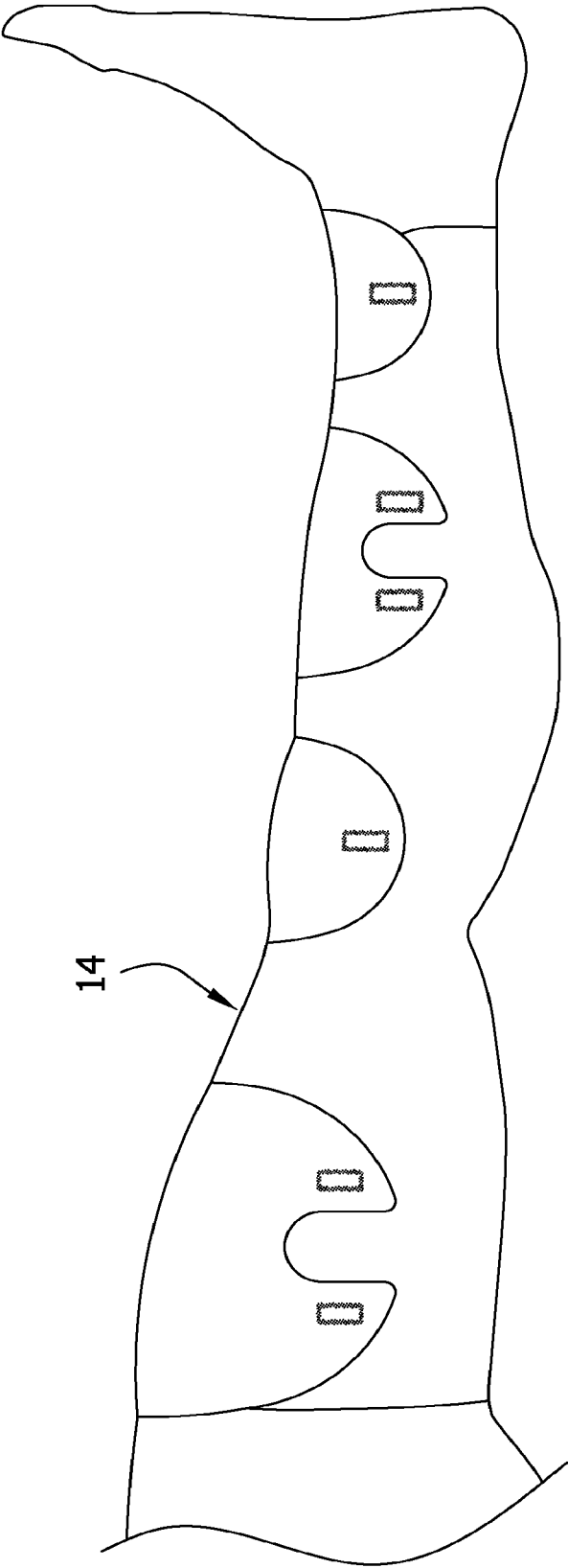


FIG. 3A

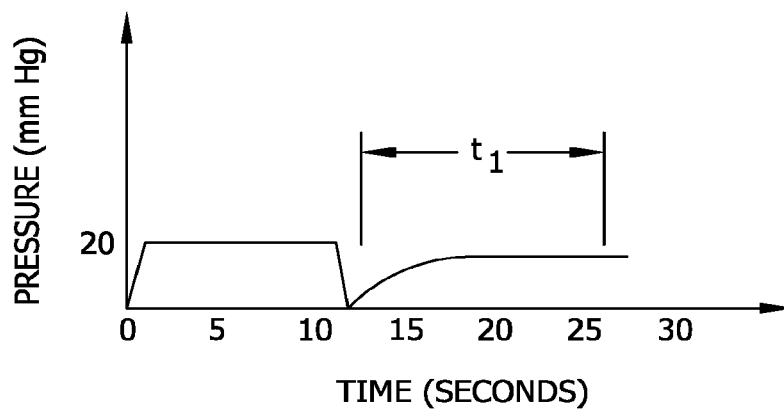


FIG. 3B

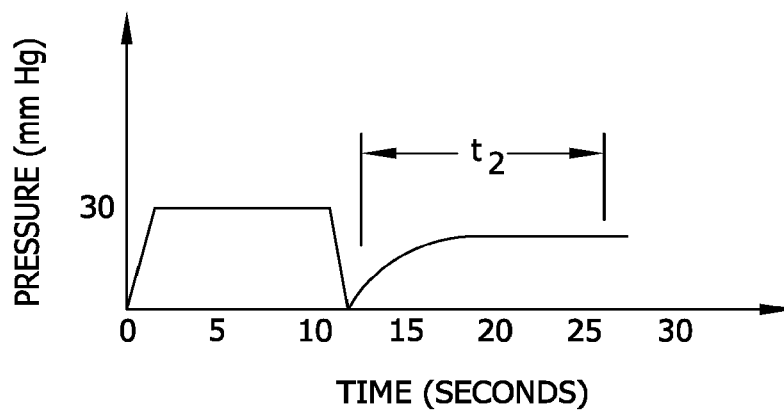


FIG. 3C

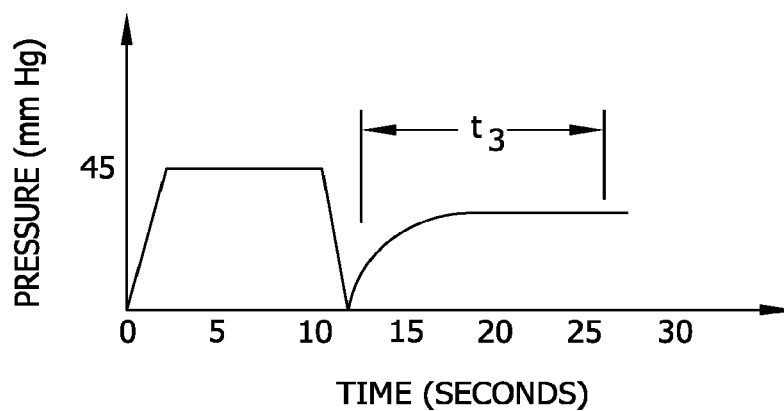


FIG. 3D

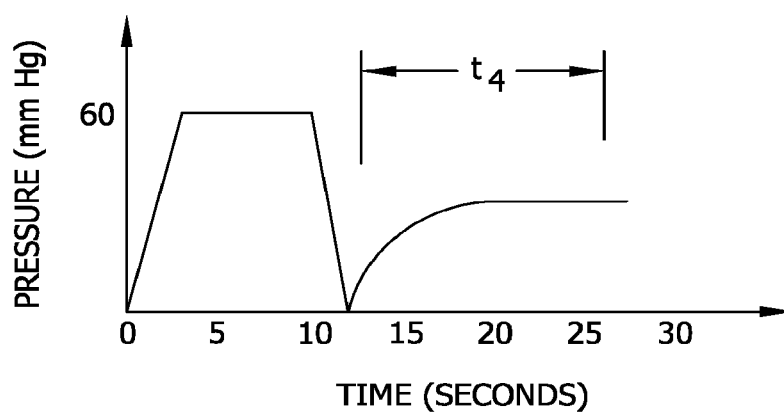


FIG. 3E

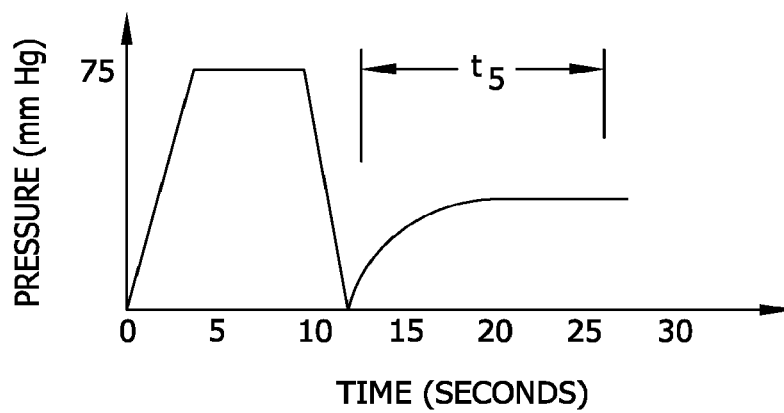


FIG. 4

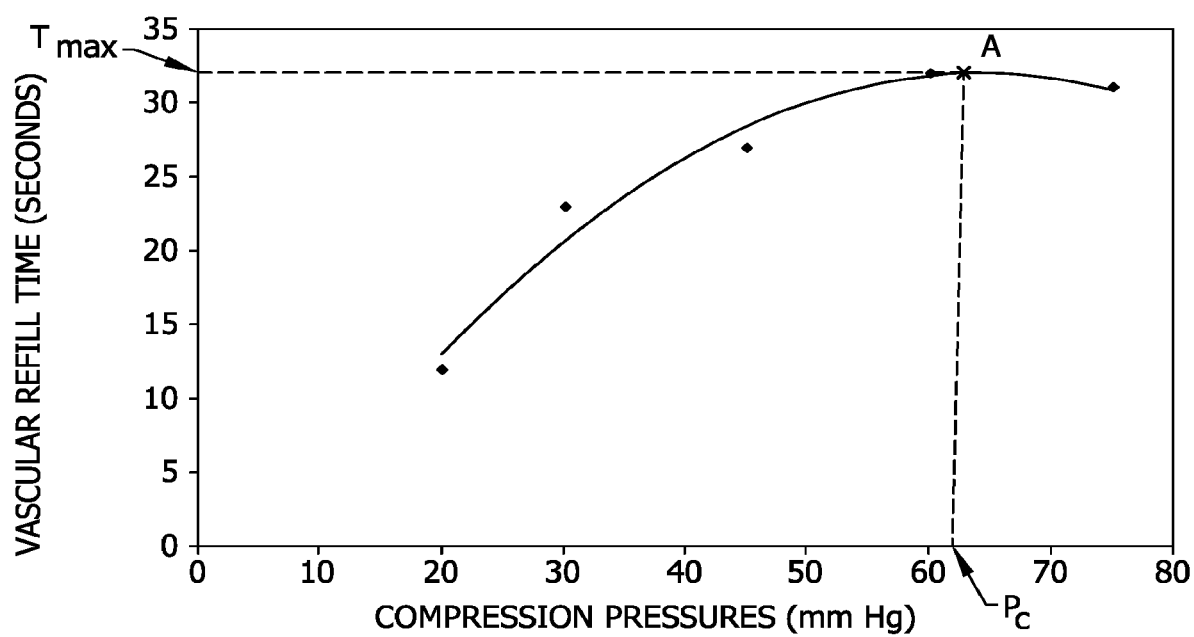


FIG. 5A

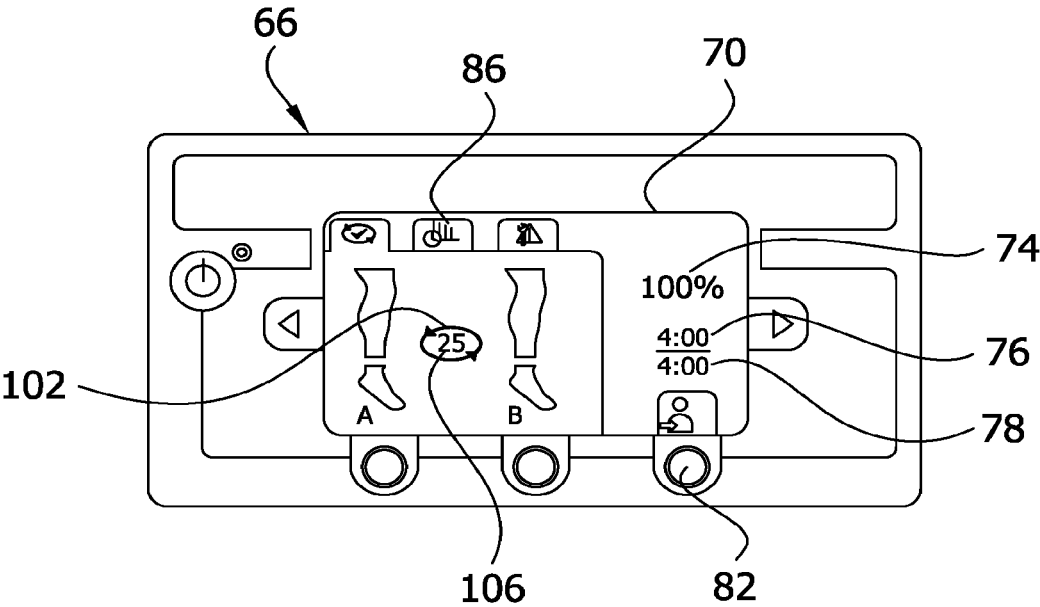


FIG. 5B

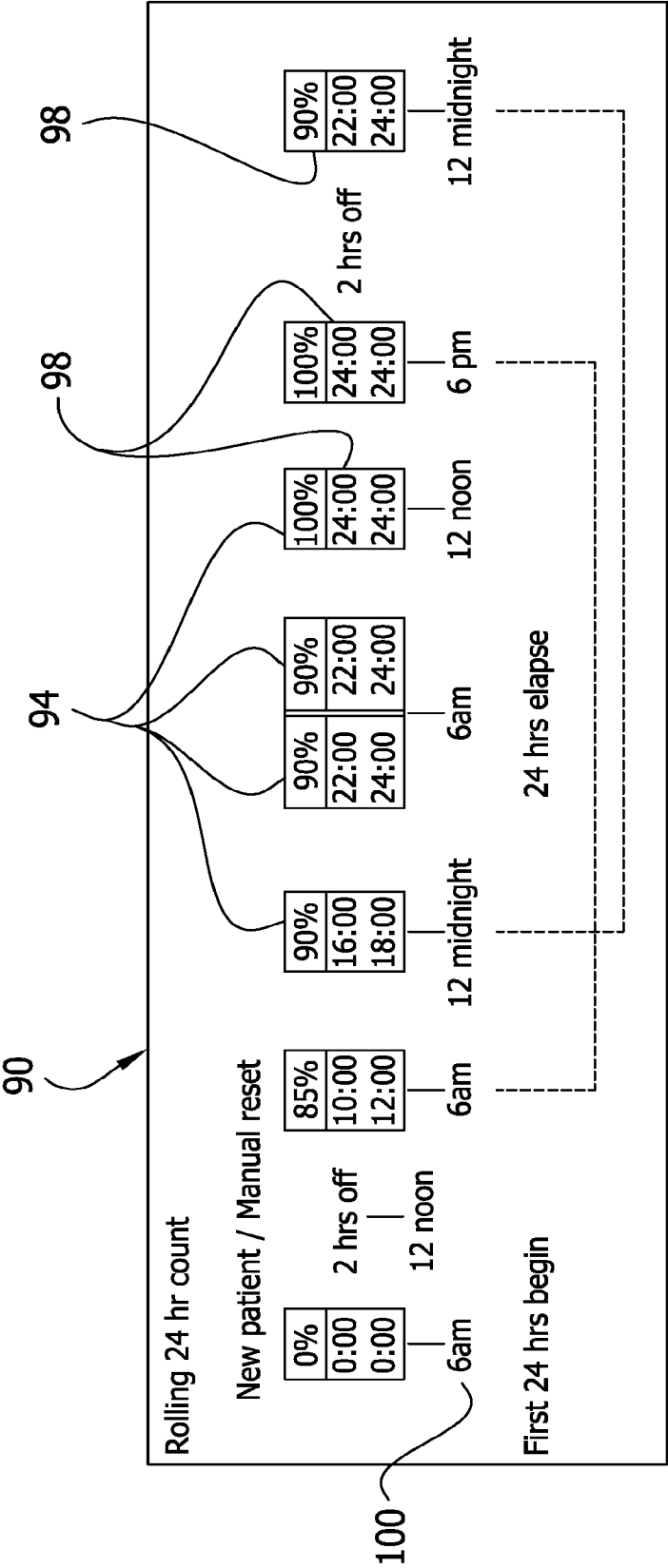


FIG. 6

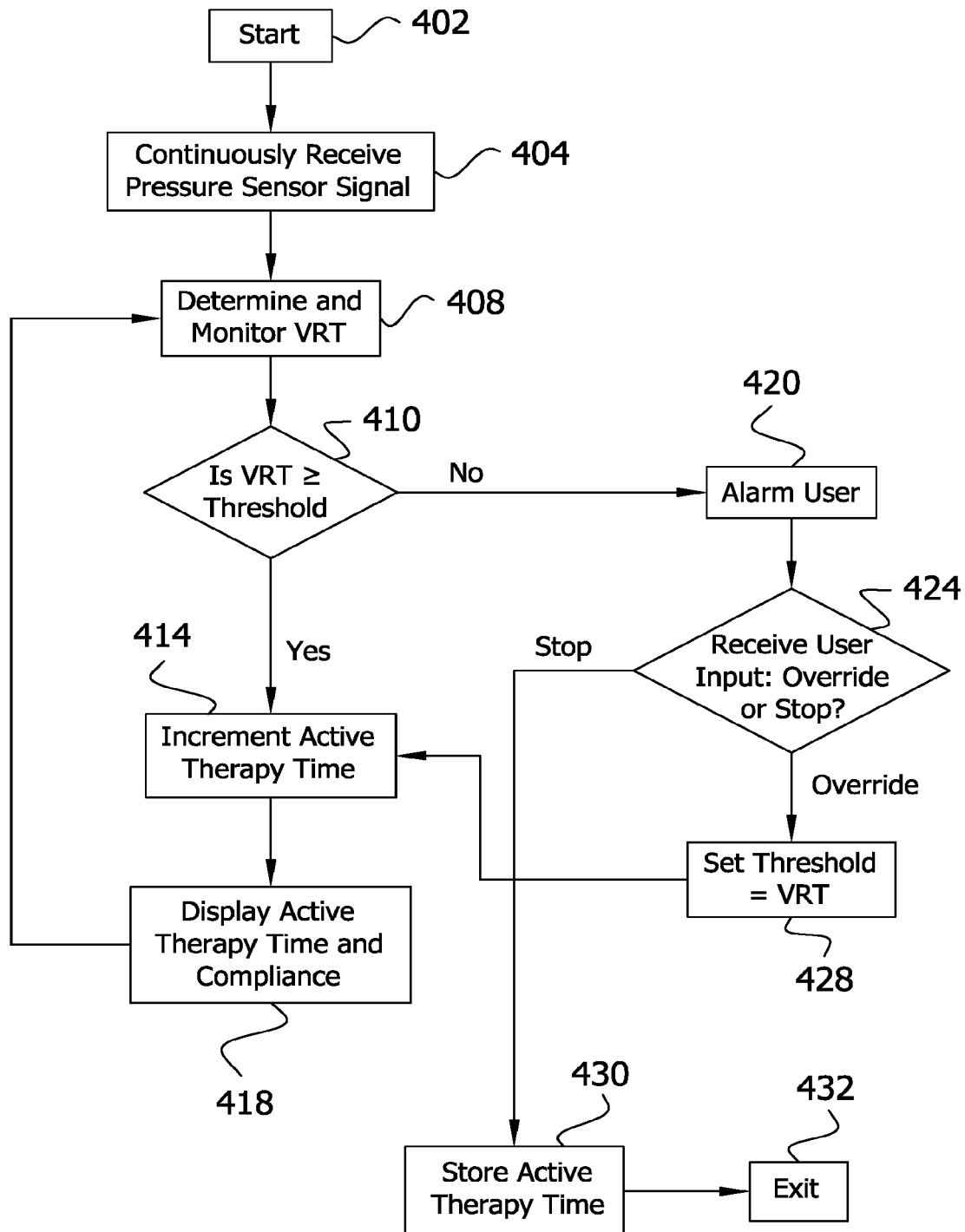


FIG. 7

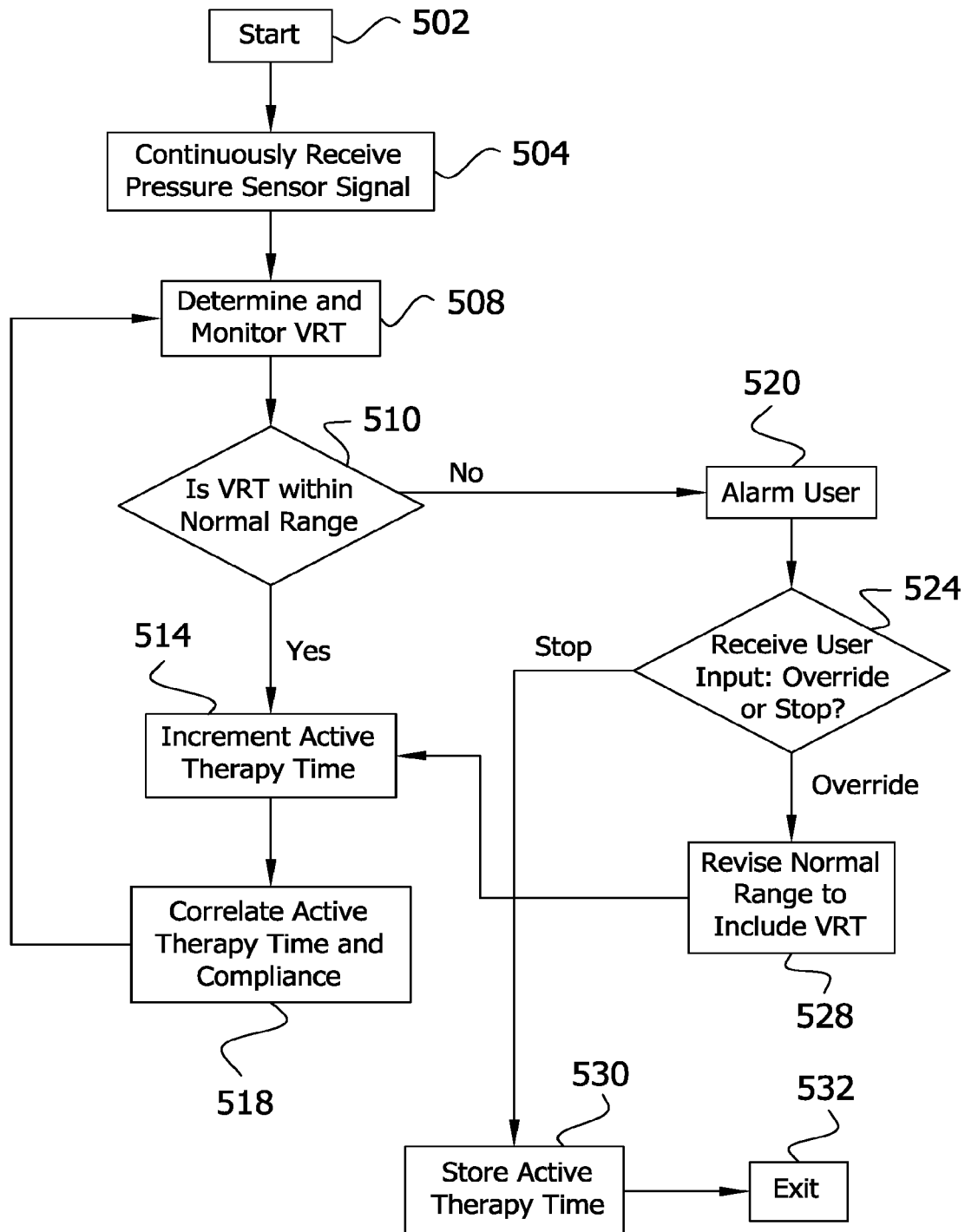


FIG. 8A

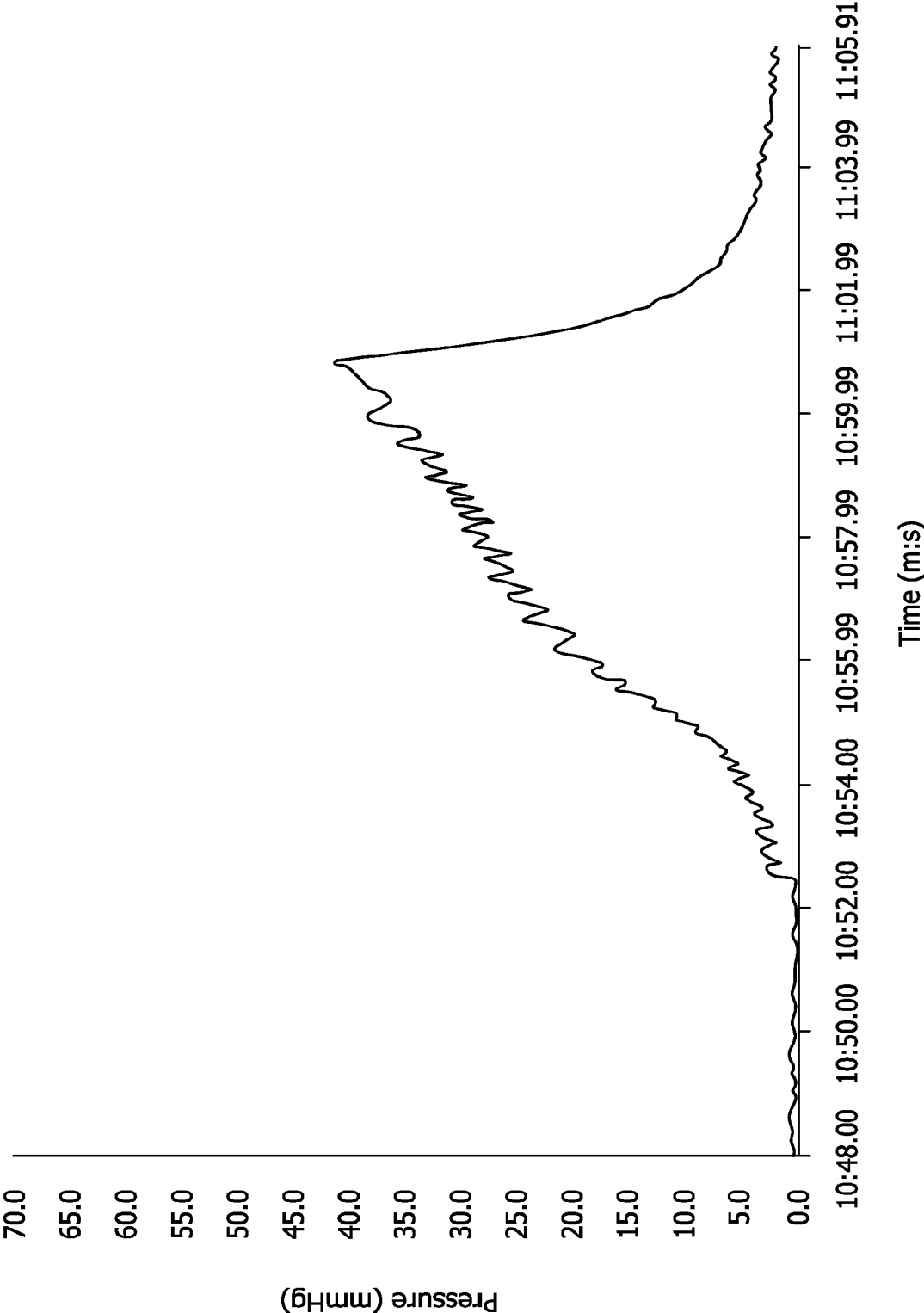


FIG. 8B

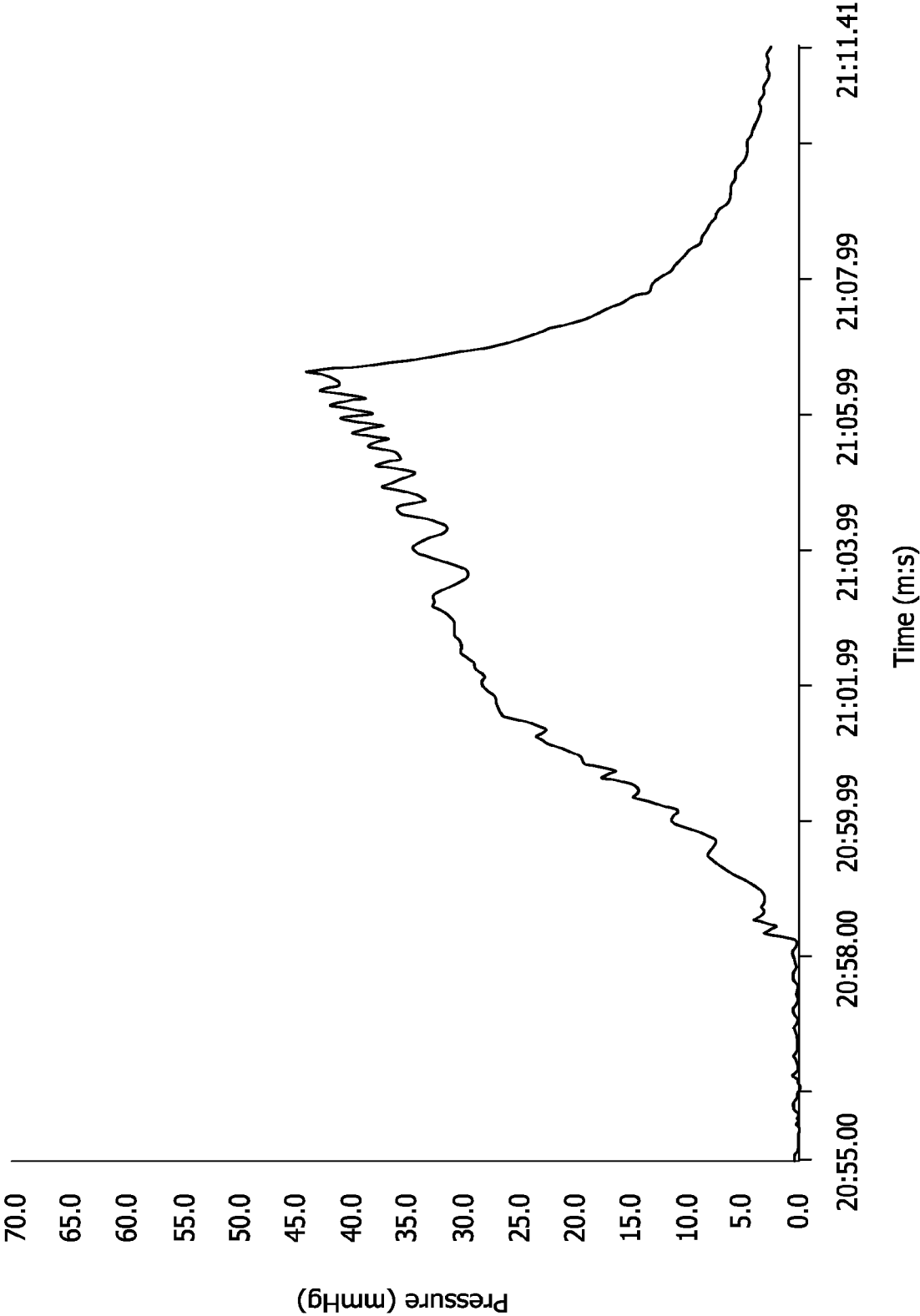
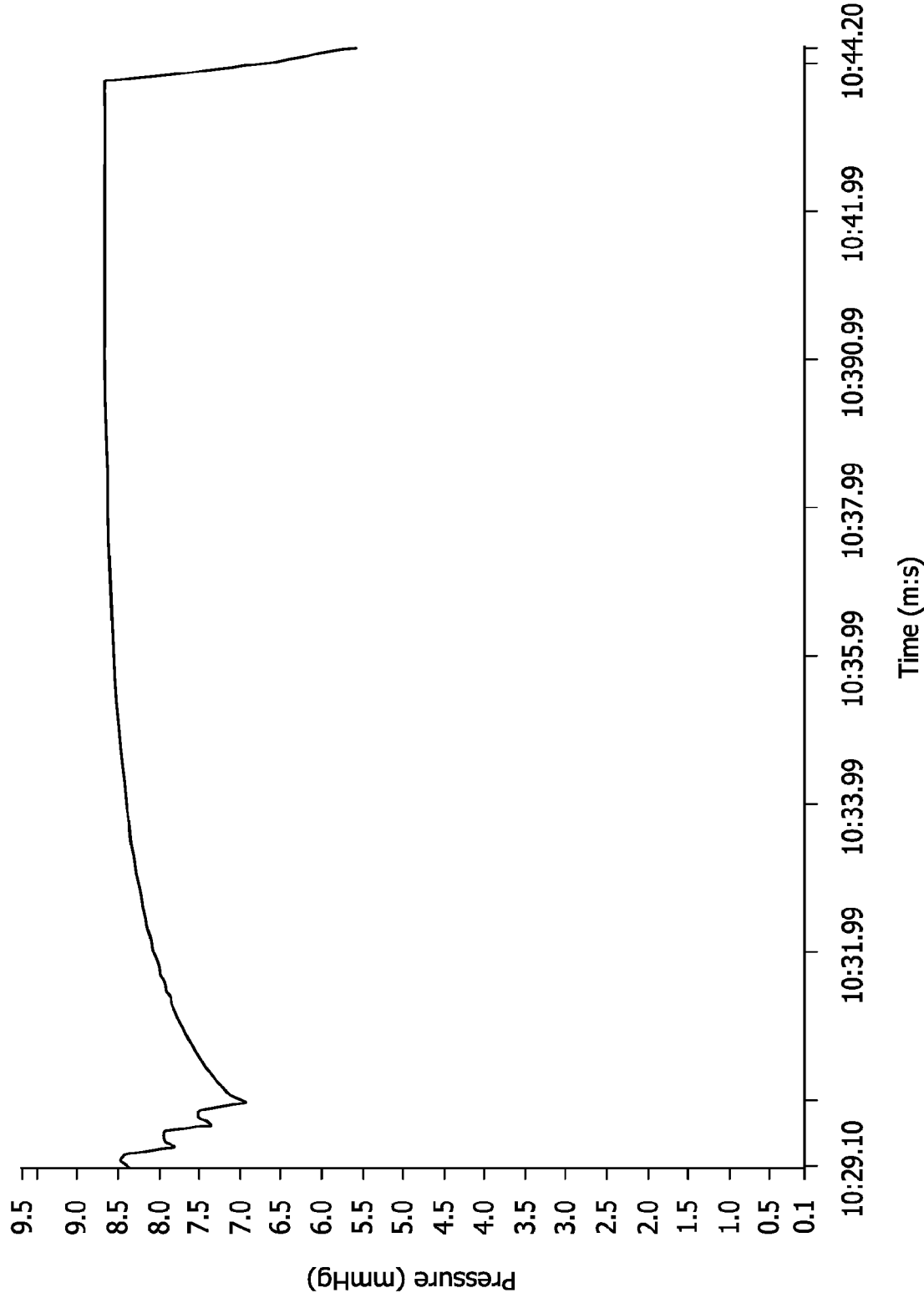
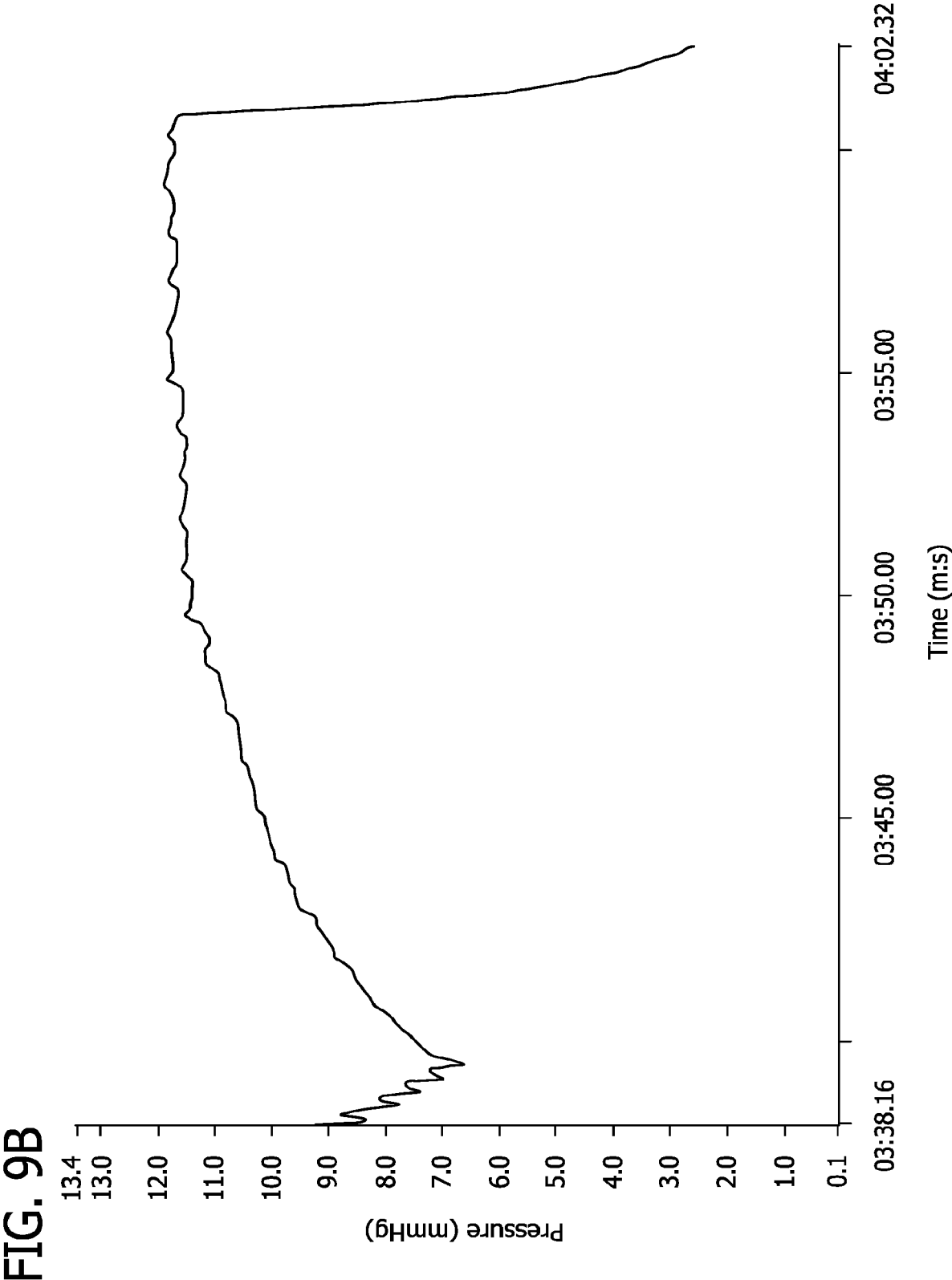


FIG. 9A





MONITORING COMPLIANCE USING VENOUS REFILL DETECTION

BACKGROUND

[0001] Aspects of the present invention generally relate to compression garments, and particularly to monitoring use of compression garments.

[0002] A major concern for immobile patients and like persons are medical conditions that form clots in the blood, such as, deep vein thrombosis (DVT) and peripheral edema. Such patients and persons include those undergoing surgery, anesthesia, extended periods of bed rest, etc. These blood clotting conditions generally occur in the deep veins of the lower extremities and/or pelvis. These veins, such as the iliac, femoral, popliteal and tibial return, deoxygenated blood to the heart. For example, when blood circulation in these veins is retarded due to illness, injury or inactivity, there is a tendency for blood to accumulate or pool. A static pool of blood may lead to the formation of a blood clot. A major risk associated with this condition is interference with cardiovascular circulation. Most seriously, a fragment of the blood clot can break loose and migrate. A pulmonary emboli can form from the fragment potentially blocking a main pulmonary artery, which may be life threatening. The current invention can also be applied to the treatment of other conditions, such as lymphedema.

[0003] Conventional vascular compression systems include a compression sleeve or garment fluidly connected to a controller for cyclically inflating the sleeve. The sleeve wraps around a patient's limb and has one or more chambers, or bladders, inflated to provide compressive pulses to the limb, typically starting around the most distal portion of the limb (e.g., the ankle) and progressing sequentially toward the heart. The cyclical inflation of the compression garment enhances blood circulation and decreases the likelihood of DVT. Also, vascular compression systems may be applied to the treatment of other conditions, such as lymphedema.

[0004] An important monitoring parameter for compression systems is the venous refilling time (VRT) calculated by the controller, which is the normal time taken for the veins in the limb to distend with blood after compression. Current devices, such as those disclosed in U.S. Pat. 6,231,532, detect pressure change (e.g., via a pressure sensor) in the sleeve as a function of the change in girth of the limb to measure VRT. In turn, the controller adjusts the cycle of compressive pulses accordingly based on the calculated VRT.

[0005] Patient compliance with a prescribed compression regimen and usage of a compression system is a common problem. Unfortunately, it is nearly impossible in a health service setting for a medical professional to constantly monitor a patient during use of the system. Therefore a need exists for improved compliance monitoring.

SUMMARY

[0006] In general, aspects of the invention relate monitoring a patient's compliance with a compression therapy regimen based on a determined VRT. In one aspect, a signal is received from a pressure sensor coupled to a compression garment that is sized and shaped to be wrapped around substantially a limb of a wearer. The signal is indicative of a change of girth of the limb. A venous refill time of the

limb is determined as a function of the received signal and monitored. When the monitored venous refill time exceeds a predetermined threshold, a patient compliance timer is incremented.

[0007] A system embodying aspects of the invention monitors patient compliance with a compression therapy regimen. The system includes a compression garment, a compression control unit, and a pressure sensor. The garment is sized and shaped to be wrapped around substantially a body part of a wearer and has one or more fasteners for use in securing the garment in a self-retaining wrapped configuration around the body part. And the garment comprises one or more selectively inflatable bladders for applying compression to the body part upon inflation. The compression control unit comprises a pump for pressurizing fluid and an outlet port in fluid communication with the pump. The outlet port has fluid tubing connected thereto for selectively delivering pressurized fluid to at least one of the inflatable bladders. The pressure sensor is coupled to at least one of the bladders and generates a signal indicative of a change of girth of the body part when the garment is in the wrapped configuration. The control unit also includes one or more processors receiving and responsive to the signal generated by the pressure sensor for determining a venous refill time of the body part. The processor monitors the determined venous refill time and increments a patient compliance timer in response to the monitored venous refill time exceeding a predetermined threshold.

[0008] In another aspect, a method of monitoring patient compliance with a compression therapy regimen includes receiving a signal from a pressure sensor coupled to a compression garment. The signal is indicative of a change of girth of a limb when a compression garment is wrapped substantially around the limb. The method includes determining a venous refill time of the limb as a function of the received signal and monitoring the determined venous refill time. The monitored venous refill time is compared to a predetermined range of normal venous refill times. The method also includes correlating the monitored venous refill time to determine patient compliance as a function of the comparing.

[0009] Other objects and features will be in part apparent and in part pointed out hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a schematic of a system for monitoring patient compliance with a compression therapy regimen;

[0011] FIG. 2A is a front view of a compression garment in an unwrapped configuration;

[0012] FIG. 2B illustrates the compression garment of FIG. 2A in a wrapped configuration adapted for use on a patient;

[0013] FIGS. 3A-3E are graphs illustrating exemplary pressure profiles during a procedure to determine venous refill time according to the present invention;

[0014] FIG. 4 is a graph illustrating an exemplary customized venous refill determination based on the pressure profiles in FIGS. 3A-3E;

[0015] FIG. 5A is an interface of the control unit according to an embodiment of the invention;

[0016] FIG. 5B is an exemplary display of patient compliance according to an embodiment of the invention;

[0017] FIG. 6 is an exemplary flowchart for monitoring compliance according to an embodiment of the invention;

[0018] FIG. 7 is an exemplary flowchart for monitoring compliance according to another embodiment of the invention;

[0019] FIG. 8A is a graph illustrating an exemplary pressure cycle of an inflatable bladder when not in use;

[0020] FIG. 8B is a graph illustrating an exemplary pressure cycle of an inflatable bladder when in use;

[0021] FIG. 9A is a graph illustrating an exemplary pressure profile during venous refill determination of an inflatable bladder when not in use; and

[0022] FIG. 9B is a graph illustrating an exemplary pressure profile during venous refill determination of an inflatable bladder when in use.

[0023] Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

[0024] Referring to FIG. 1, a compression therapy system for monitoring patient compliance is designated generally by the reference character 10. In an embodiment, system 10 is an intermittent pneumatic compression (IPC) device or the like. The system 10 comprises a garment 14 that can be fitted to a limb or other body part of a patient. The garment 14 has with one or more bladders 18a-n for applying compression to the limb during inflation of the bladders. The system 10 also includes a pressure sensor 22 coupled to at least one of the bladders (e.g., bladder 18b) via, for example, connection tubing, and a compression control unit 26. The control unit 26 monitors patient compliance with a prescribed compression therapy regimen. Specifically, and as will be described in detail later, control unit 26 determines, or calculates, a venous refill time (VRT) of the limb based on pressure measurements obtained from the pressure sensor 22. The control unit 26 monitors the determined VRT and increments an active therapy time or timer if the monitored VRT exceeds a predetermined threshold.

[0025] The garment 14, an exemplary embodiment of which is shown in FIGS. 2A-B without any interconnections to the rest of system 10, wraps substantially around a limb or body part of the patient. The garment 14 has one or more positioning or fitting devices, such as fasteners 30A-F, 32A-F, for securing the garment in a self-retaining, wrapped configuration around the limb. Desirably, the garment 14 is sized and shaped to fit the limb in a manner that avoids wasted energy typically associated with inflating a loosely fitted garment. Any suitable approach of determining fit of the garment 14 and accordingly adjusting the fasteners 30A-F, 32A-F is within the scope of the invention. Such approaches may include, but are not limited to, simple user measurements such as inserting a finger between the garment 14 and the limb to check for fit, and more complex, sensor-based fitting mechanisms formed on the garment itself.

[0026] FIG. 2A illustrates the fasteners as hook (30A-F) and loop (32A-F) tabs for this purpose. FIG. 2B shows the garment 14 in a wrapped configuration on a leg of a patient, and illustrates the fasteners 30A-F, 32A-F during use. Alternatively, buckles and/or hook and loop wraps may be employed. Any design of the fasteners 30A-F, 32A-F is within the scope of the invention. In the exemplary embodiment illustrated in FIGS. 2A and 2B, garment 14 comprises

four inflatable bladders 36-39 (referred to as bladders 18a-n in FIG. 1) for selectively applying compression to the limb upon inflation. Any number, shape, and configuration of the inflatable bladders 36-39 is within the scope of the invention.

[0027] Referring again to FIG. 1, the compression control unit 26 is operable for controlling operation of the compression system 10. A pump 42 of the unit 26 connects to a fluid supply 46 and provides a fluid (e.g., compressed air) to the bladders 18a-n via connection tubing. Specifically, an outlet port 50 of the pump 46 controls fluid delivery to the bladders 18a-n. As is typically known for IPC systems, bladders 18a-n undergo alternate inflation and deflation cycles to provide intermittent compression. The control unit 26 also comprises a processor 54 for monitoring VRT and determining patient compliance (described below), though more than one processor may be employed without departing from the scope of the invention. Processor 54 in general is operable to execute the various functions of the compression control unit described above and hereafter. For example, processor 54 executes software instructions for monitoring sensor 22 and determining VRT and for incrementing the active therapy time accordingly. Moreover, processor 54 is further configured for controlling operation of pump 42 and port 50 during operation. The pressure transducer or sensor 22 is coupled via connection tubing to one of the bladders, bladder 18b in the illustrated embodiment, for monitoring pressure in the bladder 18b. Sensor 22 is preferably coupled to port 50, and in turn coupled to bladder 18b via the same connection tubing as used by pump 42. Alternative connection means are possible as well. The monitored pressure may be employed to determine venous refill time, or VRT, of the limb during a VRT mode of the control unit 26. Referring to FIGS. 3A-3E, processor 54 is configured to execute computer-executable instructions for pressurizing the bladder 18b, for example, to determine a customized venous refill time for the bladder. In an embodiment, when it is desired to determine the venous refill time for the patient, control unit 26 permits bladder 18b to reach a compression pressure and then causes it to depressurize until the pressure in that particular bladder reaches a lower value. The computer-executable instructions for determining the venous refill time comprise pressurizing the bladder 18b to a first compression pressure (e.g., 20 mm Hg) to move the blood in the leg from a region (e.g., calf) underlying the bladder. After pressurizing the bladder 18b to the first compression pressure, the pressure in the bladder is reduced to a refill pressure (e.g., 10 mm Hg) to allow the blood to reenter the region of the limb underlying the bladder (after approximately 2.5 seconds of depressurization).

[0028] The pressure in the bladder 18b is then sensed by the pressure transducer 22 until it is determined that blood flow has been completely restored to the region of the limb underlying the bladder. The time elapsed to restore blood flow is characterized as a first venous refill time t_1 and is stored by the controller 26. The bladder 18b is then pressurized to a second compression pressure (e.g., 30 mm Hg) and the same process is performed as was performed for the first compression pressure, resulting in a second venous refill time t_2 . The bladder 18b can then be pressurized to even more compression pressures (e.g., 45, 60 and 75 mm Hg) and the process performed for the first and second compression pressures can be repeated for each pressure level to produce venous refill times t_3 , t_4 , t_5 , t_n for each additional

pressure level. It is understood that pressure amounts other than those described above and shown in FIGS. 3A-3E can be used in the venous refill process without departing from the scope of the invention. Additionally, the venous refill process at each pressure level can be performed multiple times to produce multiple venous refill times for each pressure level.

[0029] Alternatively, the bladder under inspection could be permitted to depressurize for a predetermined period of time, or to depressurize fully and then be repressurized until the pressure reaches the predetermined value, for example, 10 mm Hg. The pressure transducer 22 senses the pressure in bladder 18b for a time sufficient to allow the venous system in the limb to refill, i.e., engorge with blood again. The pressure as sensed by pressure transducer 22 rises as the limb expands upon filling with blood and reaches a generally steady state when the leg has refilled. The time between the start of depressurizing the bladder 18b and when this plateau occurs is measured to be the VRT.

[0030] For example, using the determined venous refill times t_1 - t_n , the processor 54 determines a customized compression pressure by plotting the venous refill times for each selected pressure level on a graph as shown in FIG. 4 and fitting a best fit line to the plot using standard linear regression analysis. The apex A of the best fit line corresponds to a customized compression pressure P_c for producing a maximum venous refill time T_{max} . The determined compression level P_c and refill time T_{max} are then incorporated into the compression therapy of the limb wherein the bladder 18b in the garment, or sleeve, 14 is repeatedly pressurized to the customized compression pressure P_c , maintained at the customized compression pressure for a period of time and subsequently reduced to the refill pressure for the determined maximum refill time T_{max} to facilitate blood circulation in the limb.

[0031] In the instance where multiple venous refill times are recorded for each selected compression pressure level, the refill times are averaged by the processor 54 to produce an average value for the given pressure level. These average values are then plotted and a best fit line is fit to the plot of the average values and the customized compression pressure and maximum venous refill time are extrapolated from the plot in the same manner as described above. If the garment 14 includes multiple bladders (e.g., ankle, calf and thigh bladders as shown in FIGS. 2A and 2B), the controller 26 can be configured to operate the IPC device 10 to apply sequential compression therapy to the limb using the customized pressure and maximum refill time.

[0032] In an additional or alternative embodiment, each time control unit 26 determines VRT, it cycles (i.e., inflates and deflates) bladder 18b through several values of compression pressure to obtain a corresponding VRT value for each value of compression pressure. The control unit 26 then calculates a maximum VRT, or V_{max} . V_{max} is ascertained by determining a best fit between the compression pressure values and the corresponding VRT values via any suitable fitting method (e.g., linear regression analysis). Specifically, a maxima of the best fit designated as V_{max} . Desirably, instead of using individual VRT values, multiple VRTs are recorded and averaged for each compression pressure to provide an average VRT value for each compression pressure value.

[0033] A custom compression pressure P_c is then determined corresponding to V_{max} and is designated as a target

compression pressure of the compression therapy regimen of bladder 18b.

[0034] After applying compression therapy to the limb for a period of time the process for determining the customized compression pressure and maximum venous refill time can be repeated to determine new values. Additionally or alternatively, memory in the controller 26 can record the venous refill times sensed by the pressure transducer 22 during the compression therapy and, for example, average the recorded values to adjust the time between consecutive pressurizations of the bladder 18b based on the averaged refill times. These two processes ensure that the compression therapy being delivered to the limb adapts to the changing characteristics of the limb so that a customized compression therapy is delivered to the limb through the duration of the compression therapy.

[0035] As described above, processor 54 of the control unit 26 is responsive to the output signal of pressure sensor 22 for determining the VRT as described above. The unit 26 is further operable to monitor the determined VRT over time. Any aspect of the measured VRT may be monitored, including, but not limited to: individual VRT values, average VRT within a specific time window, average VRT within a moving time window, variations in VRT over multiple VRT measurements and/or compressive cycles, the steady state pressure achieved during the VRT measurement, any compression cycle parameter, and so on.

[0036] Most patients have a normal VRT between 40-50 seconds for leg measurements, with inanimate leg forms generating VRT values as low as 30 seconds. A VRT of approximately 30 seconds is also typically observed when the garment 14 is not in use by the patient. Hence, the monitored VRT may be used for determining whether the patient is using the garment 14. Accordingly, in a preferred embodiment, control unit 26 stores and increments an active therapy time when the monitored VRT either falls within a normal range (e.g., 30-60 seconds), or simply exceeds a predetermined threshold (e.g., 30 seconds), both of which are indicative of normal usage of compression system 10. In this manner, the value of active therapy time is a measure of the patient wearing garment 14 and its sequential inflation and deflation. Alternatively, active therapy time is the cumulative time of controller operation during which the patient is deemed compliant.

[0037] In another embodiment, control unit 26 comprises an alarm 58 indicating to a user when the monitored VRT falls below the predetermined threshold. At this point, processor 54 ceases incrementing the active therapy time until further action is taken. The alarm 58 may be one or more of an audio alarm and a visual alarm. The user, typically the patient or a clinician monitoring the patient, may respond to the alarm 58 by indicating that the patient is indeed compliant, such as the case where a patient changes positions and causes an intermittent dip in monitored VRT. In other words, the user overrides the alarm. The therapy time would then continue to be incremented.

[0038] When the clinician indicates continued compliance by overriding the alarm 58 triggered by a lower VRT value (than the predetermined threshold), control unit 26 resets or revises the predetermined threshold value to the lower VRT value measured at the time of the override. In this manner, alarm 58 will not be triggered again until the monitored VRT dips to the revised threshold value. This prevents alarm 58 from becoming bothersome in the event the patient

has or often achieves a lower VRT value for a justifiable reason such as unique physiology, posture, etc.

[0039] Alternatively, in response to alarm 58, the clinician may determine that the patient is not wearing the garment 14 and is therefore not being compliant with the compression regimen. The clinician may respond by turning off control unit 26, at which point the therapy time ceases to increment. The therapy time may advantageously be stored in a memory 62, external or internal to processor 54, for continued measurement the next time the control unit 26 is started.

[0040] In yet another embodiment, control unit 26 has a configurable option that allows therapy time to continue to increment despite the monitored VRT falling below the predetermined threshold. In this embodiment, accumulation of therapy time is halted only when a clinician turns off the control unit 26, in response to alarm 58 or otherwise. Continuing to increment the therapy timer in this manner permits the clinician to closely track an operation time of control unit 26, referred to hereafter simply as controller operation time. This embodiment is beneficial when monitoring patients with uncharacteristically low VRT, such as those suffering from venous insufficiency, for example. In such a patient, low VRT measurements may erroneously indicate non-compliance during use. The clinician with knowledge of the patient's condition can then manually control accumulation of therapy time.

[0041] Determining patient compliance from active therapy time may be carried out in a number of ways. In one embodiment, patient compliance is simply the therapy time value. In another embodiment, patient compliance is specified as a ratio between active therapy time and controller operation time.

[0042] In another embodiment, a shift time is monitored and has a specified value, such as 24 hours. Compliance is specified as a ratio between active therapy time and shift time. Once monitoring is initiated, both active therapy time and shift time are continually evaluated. When the operation time of the controller reaches the shift time (i.e., operation time = 24 hours), the compliance measurement is limited to a rolling 24-hour (shift time) window. At any time point thereafter, active therapy time and hence compliance is accounted for only over the last 24 hours of operation. Desirably, shift time is programmable and resettable by a user. In this manner, a clinician or other healthcare provider can specify his or her own shift time, and then observe how long the patient has been compliant during the shift.

[0043] The control unit 26 further includes a controller interface 66. A display 70 of the interface 66, as illustrated in FIG. 5A, displays patient compliance as a percentage 74, wherein the percentage is evaluated as a ratio between the displayed active therapy time 76 and the displayed shift time 78. The interface 66 includes a RESET option 82 for resetting the timers. The display 70 also illustrates a VRT indicator 102 and a VRT value 106.

[0044] A user may further access a Compliance Graph 90 (see FIG. 5B) via a graph option 86 of the interface 66. Specifically, FIG. 5B illustrates a rolling 24-hour window for monitoring compliance and shows a percentage compliance 94. The exemplary user interface of FIG. 5B displays the percentage compliance 94 along with a boxed representation of the therapy time and shift time (denoted together by the reference character 98) at various time points. The timers are reset at time point 100, and monitored thereafter. In this example, compliance is approximately 90% for the

first 24 hours (6am-6am), 100% for the 6pm-6pm slot, and 90% for the 12 midnight-12 midnight slot. Other means of displaying compliance and the various timers are within the scope of the invention.

[0045] According to aspects of the invention, a method of monitoring patient compliance is generally illustrated in FIG. 6 in the form of an exemplary flow diagram. Compliance monitoring is initiated or reset at 402. At 404, a signal is continuously received from the pressure sensor 22 coupled to the bladder 18b. The signal is a function of bladder pressure, and is further indicative of a change of girth of the limb or body part of the patient. The venous refill time or VRT of the limb is determined and monitored as a function of the received signal at 408. At 410, a determination is made whether the monitored VRT exceeds the predetermined threshold. If this is the case, the active therapy time is incremented at 414. The active therapy time and compliance is displayed to the user at 418.

[0046] If, at 410, the monitored VRT does not exceed the threshold, the alarm 58 is initiated at 420. At 424, the user responds by either overriding the alarm 58 or stopping the control unit 26. If the user chooses to override the alarm, 58, the threshold is set to the monitored VRT value at 428, and the active therapy time continues to increment as described above. If the user chooses at 424 to stop the control unit 26, the active therapy time is stored to memory 62 at 430, and the control unit 26 shuts down at 432.

[0047] According to further aspects of the invention, a method of monitoring patient compliance is generally illustrated in FIG. 7 in the form of an exemplary flow diagram. Compliance monitoring is initiated or reset at 502. At 504, a signal is continuously received from the pressure sensor 22 coupled to the bladder 18b. The signal is a function of bladder pressure, and is further indicative of a change of girth of the limb or body part of the patient. The venous refill time or VRT of the limb is determined and monitored as a function of the received signal at 508. At 510, a determination is made whether the monitored VRT exceeds the predetermined threshold. If this is the case, the active therapy time is incremented at 514. The active therapy time is correlated to compliance, and may further be displayed to the user, at 518.

[0048] If, at 510, the monitored VRT does not exceed the threshold, the alarm 58 is initiated at 520. At 524, the user responds by either overriding the alarm 58 or stopping the control unit 26. If the user chooses to override the alarm, 58, the threshold is set to the monitored VRT value at 528, and the active therapy time continues to increment as described above. If the user chooses at 524 to stop the control unit 26, the active therapy time is stored to memory 62 at 530, and the control unit 26 shuts down at 532.

[0049] Having described aspects of the invention in detail, it will be apparent that modifications and variations are possible without departing from the scope of the invention defined in the appended claims. For example, an upper predetermined threshold (e.g., 60 seconds) of monitored VRT may be defined that triggers the alarm as well. In other words, the alarm may be triggered above and below a predetermined range of normal VRT values, typically 30-60 seconds. The upper predetermined threshold may be resettable as well. Additionally, more than one inflation bladder may be connected to a different pressure sensor each, and the pressure readings from several pressure sen-

sors may then be combined in any way possible to determine VRT and/or compliance.

[0050] The compliance percentage may, in addition to being indicated numerically as illustrated, also be displayed via graphical elements such as a pie chart (not shown). Interface **66** is desirably an integrated display with associated soft keys as illustrated, allowing the user to select and browse various elements described above using the soft keys. However, other constructions of the interface **66** are within the scope of the invention.

[0051] To improve patient compliance with compression therapy, there is a need for increasing clinician participation while providing the clinician a utility for compliance notification and monitoring. Several requirements must be fulfilled to achieve this goal. The clinician should be notified when compliance is purportedly not being achieved. Further, the clinician should be able to decide whether to deem the patient compliant or not, and adjust compliance parameters to each patient. Finally, the clinician should be able to monitor the duration of compliance for specific time periods, since they are more likely to be concerned with patient compliance during their work shift(s). A user-friendly compliance monitoring interface is provided for this purpose.

[0052] In view of the above, it will be seen that the several objects of the invention are achieved and other advantageous results attained. FIGS. **8A** and **8B** illustrate, during operation, a pressure cycle of inflatable bladder **18b**. Even when not in use by a patient, bladder **18b** reaches pressure values (see FIG. **8A**) that are similar to when the bladder **18b** is in use by a patient (see FIG. **8B**). The difference in the curve peaks is merely 2-3 mmHg for the illustrated case. When monitored by a pressure sensor, the pressure values correspond to internal bladder pressure, and cannot adequately account for actual usage of the garment **14**. Pressure profiles for measuring VRT, on the other hand, are determined by reducing bladder pressure to a refill pressure and closing a vent valve (as described above), followed by monitoring pressure increase as blood reenters the limb. Blood flow to the limb results in expansion of the limb, which forces air out of bladder **18b**, back through connecting tubing and onto the sensor **22**, which records an increase in pressure. Processor **54** then evaluates the increase in pressure to calculate VRT and determines compliance.

[0053] FIGS. **9A** and **9B** illustrate the pressure profile as a refill curve of bladder **18b** during VRT measurement. The illustrated pressure profile compares two scenarios, namely, a) when garment **14** is not in use by a patient (see FIG. **9A**) and b) during use by a patient (see FIG. **9B**). When no blood flow is detected such as during non-use, an insignificant increase in pressure is observed, a little less than 2 mmHg for the illustrated case and attributable to pressure stabilization. During use, on the other hand, a pressure change as high as 10 mmHg is observable (approximately 5.5 mmHg for FIG. **9B**) in bladder **18b** due to distension of the limb. Embodiments of the invention translate this detectable change in pressure to VRT and for indication of compliance, thereby providing a strong correlation between actual use and estimated compliance.

[0054] Additionally, by using the same pressure sensor and output to monitor VRT and usage, a controller is able to determine compliance without requiring additional, cumbersome hardware on the garment itself.

[0055] When introducing elements of the present invention or the preferred embodiments(s) thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

[0056] As various changes could be made in the above constructions, products and methods without departing from the scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

1-21. (canceled)

22. A compression garment control unit for monitoring compliance of a user with respect to wearing a compression garment in accordance with a compression treatment regimen, the control unit comprising:

a display configured to display patient compliance with a compression treatment regimen;

an alarm;

at least one computer readable storage medium configured for storing one or more monitored parameters;

one or more processors coupled with the at least one computer readable storage medium; and

computer-executable instructions embodied on the at least one computer readable storage medium, the computer-executable instructions including instructions for causing the one or more processors to:

direct a flow of fluid from a pressurized fluid flow source to cyclically inflate and deflate at least one inflatable bladder of a compression garment configured to be wrapped around a limb of a wearer of the garment;

receive pressure signals indicative of fluid pressure in the at least one inflatable bladder from a pressure sensor communicatively coupled thereto during at least one of inflation and deflation of the at least one inflatable bladder in a plurality of successive compression cycle;

process the received pressure signals; and
initiate the alarm in response to the received pressure signals indicating an interruption of operation or non-compliance with compression therapy.

23. The compression garment control unit of claim **22**, wherein the alarm comprises one or more of an audio alarm and a visual alarm.

24. The compression garment control unit of claim **22**, wherein the sensor is external to the at least one inflatable bladder.

25. The compression garment control unit of claim **22**, wherein the sensor is coupled with the at least one inflatable bladder via a connection tubing.

26. The compression garment control unit of claim **22**, wherein the at least one processor is further configured to increment an active therapy time in response to the received pressure signals indicating compliance with the compression therapy.

27. The compression garment control unit of claim **26**, wherein the at least one processor is further configured to display the active therapy time and compliance on the display in response to the received pressure signals indicating compliance with the compression therapy.

28. The compression garment control unit of claim **22**, wherein the at least one processor is further configured to increment an active therapy time in response to receiving an override command in response to the alarm.

29. The compression garment control unit of claim **28**, wherein the at least one processor is further configured to display the active therapy time and compliance on the display in response to the received override command.

30. The compression garment control unit of claim **22**, wherein the at least one processor is further configured to store an active therapy time in response to receiving a stop command in response to the alarm.

31. A compression garment system comprising:
a compression garment; and

a compression garment control unit comprising:

a display configured to display patient compliance with a compression treatment regimen;

an alarm;

at least one computer readable storage medium configured for storing one or more monitored parameters;

one or more processors coupled with the at least one computer readable storage medium; and

computer-executable instructions embodied on the at least one computer readable storage medium, the computer-executable instructions including instructions for causing the one or more processors to:

direct a flow of fluid from a pressurized fluid flow source to cyclically inflate and deflate at least one inflatable bladder of a compression garment configured to be wrapped around a limb of a wearer of the garment;

receive pressure signals indicative of fluid pressure in the at least one inflatable bladder from a pressure sensor communicatively coupled thereto during at least one of inflation and deflation of the at least one inflatable bladder in a plurality of successive compression cycle;

process the received pressure signals; and
initiate the alarm in response to the received pressure signals indicating an interruption of operation or non-compliance with compression therapy.

32. The compression garment system of claim **31**, wherein the alarm comprises one or more of an audio alarm and a visual alarm.

33. The compression garment system of claim **31**, wherein the sensor is external to the at least one inflatable bladder.

34. The compression garment system of claim **31**, wherein the sensor is coupled with the at least one inflatable bladder via a connection tubing.

35. The compression garment system of claim **31**, wherein the at least one processor is further configured to increment an active therapy time in response to the received pressure signals indicating compliance with the compression therapy.

36. The compression garment system of claim **35**, wherein the at least one processor is further configured to display the active therapy time and compliance on the display in response to the received pressure signals indicating compliance with the compression therapy.

37. The compression garment system of claim **31**, wherein the at least one processor is further configured to increment an active therapy time in response to receiving an override command in response to the alarm.

38. The compression garment system of claim **37**, wherein the at least one processor is further configured to display the active therapy time and compliance on the display in response to the received override command.

39. The compression garment system of claim **31**, wherein the at least one processor is further configured to store an active therapy time in response to receiving a stop command in response to the alarm.

40. A method for monitoring and expressing compliance with a compression device comprising steps of:

directing a flow of fluid from a pressurized fluid flow source to cyclically inflate and deflate an inflatable bladder of a compression garment configured to be wrapped around a limb of a wearer of a garment;

receiving pressure signals indicative of fluid pressure in the inflatable bladder from a pressure sensor communicatively coupled thereto during an inflation and deflation of the inflatable bladder in a plurality of successive compression cycle;

processing the received pressure signals; and

initiating an alarm in response to the received pressure signals indicating an interruption of operation or non-compliance with compression therapy.

* * * * *